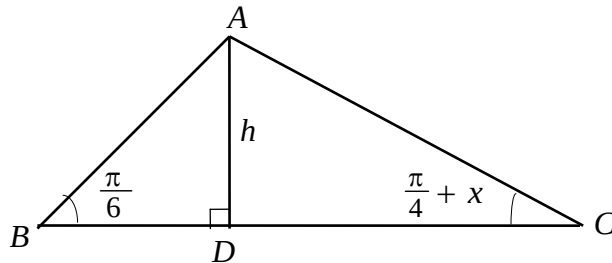


Section A: Pure Mathematics (40 marks)

1. The diagram shows triangle ABC . It is given that the height AD is h units,
 $\angle ABD = \frac{\pi}{6}$ and $\angle ACD = \frac{\pi}{4} + x$.



Show that if x is sufficiently small for x^3 and higher powers of x to be neglected, then
 $BC \approx h(1 + \sqrt{3} - 2x + 2x^2)$. [5]

2. (i) By using the substitution $u = \sqrt{x-1}$, find the exact value of $\int_1^2 x\sqrt{x-1} \, dx$. [4]

- (ii) The curve C has equation $y^2 = x\sqrt{x-1}$. The region R is bounded by the curve C , the line $y = \sqrt{2}$ and the axes. Find the exact volume of the solid of revolution when R is rotated through 2π radians about the x -axis. [2]

3. Sketch, on an Argand diagram, the locus of z which satisfies

$$|z - 4 - 3i| = 1 \text{ and } \operatorname{Re}(z) \geq 4. \quad [3]$$

Hence find

- (i) the values of a and b such that $a \leq |z| \leq b$, [3]
 (ii) the maximum and minimum values of $\arg(z - 2i)$. [2]

4. The point A has position vector $\mathbf{i} + \mathbf{j} + 3\mathbf{k}$ and the plane, π , has equation $\mathbf{r} \cdot (3\mathbf{i} + 4\mathbf{k}) = 40$.

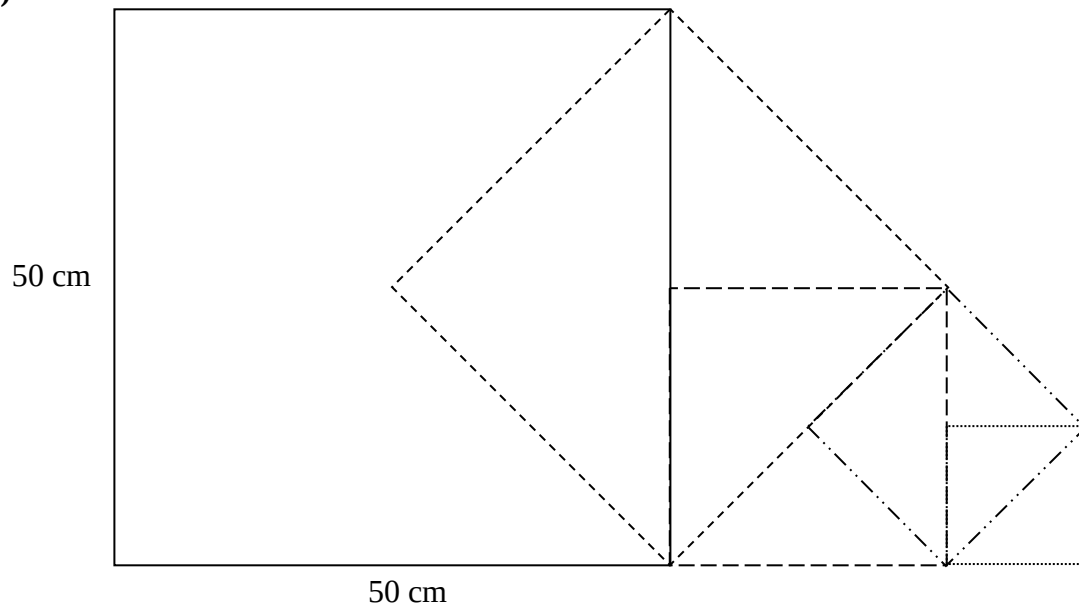
Find

- (i) the vector equation of the line l passing through A and normal to the plane, π , [1]
 (ii) the position vector of the foot of perpendicular from the point A to the plane, π . [3]

A sphere of radius 2 units has its centre at A . Show that the shortest distance between the sphere and the plane, π , is 3 units. Hence find the position vector of the nearest point on the sphere to the plane. [5]

5. (a) Show that the sum of the positive integers from $2k$ to $4k$ inclusive is $3k(2k+1)$.
 Hence, or otherwise, find the sum of the *odd* integers lying between $2k$ and $4k$. [6]

(b)



Starting with a square of side 50 cm, a series of squares is constructed with the diagonal of each square being the side of the previous one. The areas, in cm^2 , of the successive squares are denoted by $A_1, A_2, \dots, A_n, \dots$.

If $A_1 = 2500$,

- (i) find the smallest value of n for which $A_n < 1$, [4]
 (ii) evaluate $A_2 + A_4 + A_6 + A_8 + \dots$. [2]

Section B: Statistics (60 marks)

6. The student population of a primary school is made up of 6 levels as shown below. A random sample of 400 students is to be chosen from the population to investigate the average number of hours spent watching television by a student per day. State a suitable sampling method and describe in detail how this can be carried out. [4]

School Level (Primary)	Number of students
1	125
2	200
3	180
4	150
5	225
6	120
Total	1000

7. Ann and her sister Angela attend a wedding banquet with their parents. They are shown to a round 10-seater table. Assuming that the table is to be fully occupied and the seats are numbered, find the number of possible arrangements if the family is seated together, but Ann and Angela are not seated next to each other. [4]

8. A machine is designed to produce screws of length 8 millimeters. A random sample of nine screws were selected and the lengths, in millimeters, are found to be as follows:

7.99, 8.01, 8.00, 8.02, 8.03, 7.99, 8.00, 8.01, 8.01

- (a) It is required to test whether the machine is producing screws of length more than 8 millimeters.
- (i) State suitable null and alternative hypotheses, and the assumption required for this test. [2]
- (ii) Determine the least level of significance at which the test would result in the rejection of the null hypothesis. [3]
- (b) It is known that the standard deviation of the length of screws produced by the machine is 0.013 millimeters. Test, at the 10% level of significance, whether the machine is faulty. [3]

9. Singapore College wishes to determine the relationship between the Mathematics performance and Physics performance of their students. The college issued a test for each of the subjects and recorded the Mathematics score (x) and the Physics score (y) of a random sample of 7 students. The following set of data was obtained:

Math score, x	98	84	75	66	50	43	31
Physics score, y	96	91	86	85	73	66	54

- (i) Give a sketch of the scatter diagram for the data. [2]

The following models were proposed.

$$A: y = a + b \ln(x)$$

$$B: y^2 = a + bx$$

$$C: y = a + \frac{b}{x}$$

- (ii) State, with reasons, which one of the above models is the most appropriate for this set of data. [3]
- (iii) For the most appropriate model, calculate the least squares estimates of a and b . [1]
- (iv) Estimate the Physics score of a student scoring 10 marks for Mathematics using the model in (ii). Comment on the validity of this estimate. [2]
- (v) The Head of Mathematics Department used the model in (ii) to estimate the Mathematics score of a student scoring 75 marks for Physics. Comment on the suitability of this method. [1]

10. A factory produces bottled fruit juice. The empty bottles are supplied to the factory and a machine fills the bottles in succession with fruit juice. The mass M of bottles supplied to the factory is known to have normal distribution with mean 200 g and standard deviation 4.5 g. The mass L of fruit juice put into the bottles by the machine is also known to have a normal distribution, and M and L are known to be independent. The mass of filled bottles have a normal distribution with mean 1200 g and standard deviation 7.5 g.

- (i) Find the mean of L , and show that the standard deviation of L is 6 g. [2]
- (ii) Find the probability that the total mass of fruit juice put into 2 bottles exceeds one-half of the mass of a filled bottle by more than 1395 g. [3]

The machine which fills the bottles is adjusted so that the standard deviation of L is changed. Find the standard deviation of L such that the probability of L differing from its mean by no more than 500 g is 0.9. [3]

11. A salesman sells two brands of handphones. He sells on average 5.25 Kampung handphones and 8.75 Toki handphones respectively in a week. The sales of the two brands of handphones are assumed to be independent.

- (i) Find the probability that, in a randomly chosen two-week period, more than 12 handphones are sold in the first week and exactly 16 handphones are sold in the second week. **[3]**
- (ii) Find the probability that in a particular week, the salesman sells fewer than 3 Toki handphones given that he sells 12 handphones in total. **[4]**
- (iii) In a randomly chosen fourteen-week period, estimate the probability that the total number of Kampung handphones sold by the salesman exceeds 75. **[4]**

The salesman will be paid a bonus if he sells at least 15 handphones per week. If the probability that the salesman is paid a bonus for more than k weeks in the year is less than 0.05, find the least value of k . [You may assume a year has 52 weeks.] **[3]**

12. A brilliant child pronounces 1 word wrongly for every 20 words given.

- (i) Calculate the probability that in a randomly chosen book containing 100 words, he pronounces at least 2 words wrongly. **[3]**
- (ii) 3 books each containing 100 words are randomly chosen. Find the probability that the child pronounces more than 1 word wrongly from one of the books and pronounces exactly 1 word wrongly from each of the remaining books. **[3]**

50 books, each containing 100 words, are given to the child to read, estimate the probability that

- (iii) the average number of words per book that the child pronounces wrongly is between 4 and 7 inclusive, **[3]**
- (iv) there are more than 46 books of which the child pronounces at least 2 words wrongly per book. **[4]**

End of Paper